1 Original CFG Paper

Tim Salimans Jonathan Ho. "Classifier-Free Diffusion Guidance". *NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications* (2021)

Summary

- CFG extends vanilla classifier guidance to trade off mode coverage and sample fidelity in conditional diffusion model sampling in analogy to low temperature sampling or truncation in other types of generative models.

- CFG may be better than vanilla classifier guidance could be seen as an adversarial attack on classifiers, which could boost classifier-based metrics like FID and IS

- Implementing truncation or low temperature sampling in diffusion models are ineffective. Scaling model scores or decreasing the variance of Gaussian noise of the reverse process generates blurry, low quality samples (diffusion models beat GANs paper).

- Vanilla CG adapts the learned score $\boldsymbol{\epsilon}_\theta(\mathbf{z}_\lambda, \mathbf{c}) \approx -\sigma_\lambda \nabla_{\mathbf{z}_\lambda} \log p(\mathbf{z}_\lambda \mid \mathbf{c})$ to include the gradient of the log likelihood of an auxiliary classifier model $p_\theta(\mathbf{c} \mid \mathbf{z}_\lambda)$ as follows:

  $$\tilde{\boldsymbol{\epsilon}}_\theta(\mathbf{z}_\lambda, \mathbf{c}) = \boldsymbol{\epsilon}_\theta(\mathbf{z}_\lambda, \mathbf{c}) - w\sigma_\lambda \nabla_{\mathbf{z}_\lambda} \log p_\theta(\mathbf{c} \mid \mathbf{z}_\lambda) \approx -\sigma_\lambda \nabla_{\mathbf{z}_\lambda} \left[\log p(\mathbf{z}_\lambda \mid \mathbf{c}) + w \log p_\theta(\mathbf{c} \mid \mathbf{z}_\lambda)\right]$$

  This results in approximate samples from the distribution:

  $$\tilde{p}_\theta(\mathbf{z}_\lambda \mid \mathbf{c}) \propto p_\theta(\mathbf{z}_\lambda \mid \mathbf{c})\, p_\theta(\mathbf{c} \mid \mathbf{z}_\lambda)^w$$

  Results on a toy 2D dataset with 3 classes for which the conditional distribution for each class is an isotropic Gaussian show that the contional upon applying guidance is non-Gaussian. As guidance strength is increased, each conditional places probability mass farther away from other classes and towards directions of high confidence given by logistic regression, and most of the mass becomes concentrated in smaller regions.

- CFG aims to achive the same modification of the conditional score but without a classifier. It is done by training a conditional and unconditional model with one neural network and trade off both scores during sampling

  $$\tilde{\boldsymbol{\epsilon}}_\theta(\mathbf{z}_\lambda, \mathbf{c}) = (1+w)\boldsymbol{\epsilon}_\theta(\mathbf{z}_\lambda, \mathbf{c}) - w\boldsymbol{\epsilon}_\theta(\mathbf{z}_\lambda)$$

  Only a relatively small portion of the model capacity of the diffusion model needs to be dedicated to the unconditional generation task in order to produce classifier-free guided scores effective for sample quality.

Questions

- Trade-off between diversity and sample quality (truncation in GAN, temperature sampling in flows, cfg in diffusion). Intuition why these two things are opposing?
- In the 2D toy example, why is the distribution changed in such a way?
- "Furthermore, $\tilde{\boldsymbol{\epsilon}}_\theta$ is constructed from score estimates that are non-conservative vector fields due to the use of unconstrained neural networks, so there in general cannot exist a scalar potential such as a classifier log likelihood for which $\tilde{\boldsymbol{\epsilon}}_\theta$ is the classifier-guided score." What does that mean?
- So basically CFG is trained to mostly be a conditional model and then the conditional score is again heavily amplified during sampling?

2 Boosting Diffusion Models with an Adaptive Momentum Sampler

Xiyu Wang et al. "Boosting Diffusion Models with an Adaptive Momentum Sampler." *IJCAI* (2024)

Summary

- Standard Diffusion model sampling is independent of previous time steps, which can lead to excessive noise and missing high-level components in the generated images.
- By using an ADAM-like sampler and employing a larger momentum that heavily re-uses the update direction of early steps leads to augmented details while still preserving high-level information. This phenomenon aligns with the two stages of learning for likelihood-based models (Rombach et al. 2022), where the first stage focuses on learning high-frequency details, while the second stage focuses on learning conceptual compositions.

Questions

- "IS (...) is often referred to as a method to measure inter-class diversity and is less sensitive to the diversity of the images inside one label (Lucic et al. 2018; Borji 2019). In contrast, FID can detect intra-class mode collapsing as well (Lucic et al. 2018)." Does that mean that the blind spot of both metrics are non diverse intra-class samples which are however not a mode collapse? Is that even desired?
- The authors show a rate (bits/dim) vs distortion (RMSE) curve. What does that mean?

3 Classifier-Free Guidance is a Predictor-Corrector

Arwen Bradley and Preetum Nakkiran. "Classifier-Free Guidance is a Predictor-Corrector". *arXiv preprint arXiv:2408.09000* (2024)

Summary

- CFG interacts differently with DDIM and DDPM, and neither sampler with CFG generates the gamma powered distribution $p(x \mid c)^{\gamma} p(x)^{1-\gamma}$
- CFG can be interpreted as a predictor-corrector methods that alternates between denoising and sharpening
- In the SDE limit, CFG is equivalent to combining a DDIM predictor for the conditional distribution with a Langevin dynamics corrector for the gamma-powered distribution $p_t(x)^{1-\gamma'} p_t(x \mid c)^{\gamma'}$, with $\gamma' = (2\gamma - 1)$
- Understanding CFG (p^* as the ideal CFG sampler, $\hat{p}$ as the real CFG sampler) - Why CFG improves both image quality and promt adherence compared to conditional sampling?:

$$\underbrace{\text{PerceivedQuality } [\widehat{p}_\gamma]}_{\text{Real CFG}} = \underbrace{\text{PerceivedQuality } [p^*_\gamma]}_{\text{Ideal CFG}} - \underbrace{\Big(\text{PerceivedQuality } [p^*_\gamma] - \text{ PerceivedQuality } [\widehat{p}_\gamma] \Big)}_{\text{Generalization Gap}}$$

  Which means at least one of the following occurs:

  - The ideal CFG sampler improves in quality with increasing γ. That is, CFG distorts the population distribution in a favorable way (e.g. by sharpening it, or otherwise).
  - The generalization gap decreases with increasing γ. That is, CFG has a type of regularization effect, bringing population and empirical processes closer.

  It is likely that both occurs: most models are trained on poor-quality datasets (cluttered images from the web) and we want the model to sample from a higher-quality distribution (images of an isolated subject). On the other hand, CFG seems to produce less outliers.

- It is open to find explicit characterizations of CFGs output distribution, in terms of the original $p(x)$ and $p(x|c)$ — although it is possible tractable expressions do not exist.

Questions

- "We cannot mathematically specify this notion of quality, but we will assume it exists for analysis. Notably, PerceivedQuality is not a measurement of how close a distribution is to the ground-truth p(x—c) — it is possible for a generated distribution to appear even "higher quality" than the ground-truth, for example." How?

- For the case of the reduced generalization gap, is that somehow related to the gap in the ELBO?

4 What does guidance do? A fine-grained analysis in a simple setting

Muthu Chidambaram et al. "What does guidance do? A fine-grained analysis in a simple setting". *NeurIPS* (2024)

Summary

- The paper analyzes two cases: (1) mixtures of compactly supported distributions and (2) mixtures of Gaussians, which reflect salient properties of guidance that manifest on real-world data.
- As the guidance parameter increases, the guided model samples more heavily from the boundary of the support of the conditional distribution. The guided ODE first swings past the edges of the support of p and into the tails of the noised data distribution p_t before returning. As a result, errors in score estimation at these tails can move the sampling process away from the intended trajectory and thus prevent the trajectory from ever returning to the support of p, leading to corrupted outputs.
- For any nonzero level of score estimation error, sufficiently large guidance will result in sampling away from the support, theoretically justifying the empirical finding that large guidance results in distorted generations.

5 Guiding a Diffusion Model with a Bad Version of Itself

Tero Karras et al. "Guiding a diffusion model with a bad version of itself". *NeurIPS* (2024)

Summary

- CFG uses an unconditional model to guide a conditional model, leading to simultaneously better prompt alignment and higher-quality images at the cost of reduced variation. Interestingly, it is possible to obtain disentangled control over image quality without compromising the amount of variation by guiding generation using a smaller, less-trained version of the model itself rather than an unconditional model.
- CFG forces the model to generate samples away from other classes, which can mean dropping density areas close to another class. CFG also avoids low-probability intermediate regions which can mean gaps in the density

where CFG wont sample from. The authors say this may stem from a worse fit of the data distribution for the unconditional model due to the fact that it is much more difficult to generate out of the entire distribution, paired with the fact that the unconditional model usually gets a small part of the training budget. This could mean CFG is not only boosting the likelihood of the sample having come from the target class, but also that of having come from the higher-quality implied distribution.

- Autoguidance uses this intuition to guide a high-quality model with a poor quality model, usually with low capacity or under-trained. This is because under limited model capacity, score matching tends to over-emphasize low-probability (i.e., implausible and under-trained) regions of the data distribution. The authors expect the weaker version of the same model to make broadly similar errors in the same regions, only stronger. Measuring the difference between both scores indicates the general direction towards better samples if the models disagree. The authors recommend an early snapshot of a smaller model as the guidance model. Many other degradation models to yield a guidance model are not performing well.

Questions

- ”The training objective of a diffusion model aims to cover the entire (conditional) data distribution. This causes problems in low-probability regions: The model gets heavily penalized for not representing them, but it does not have enough data to learn to generate good images corresponding to them. CFG has become the standard method for “lowering the sampling temperature”, i.e., focusing the generation on well-learned high-probability regions. By training a denoiser network to operate in both conditional and unconditional setting, the sampling process can be steered away from the unconditional result — in effect, the unconditional generation task specifies a result to avoid.”. Should the training objective then not be fixed to ignore low probability regions from the start?

6 Applying Guidance in a Limited Interval Improves Sample and Distribution Quality in Diffusion Models

Tuomas Kynkäänniemi et al. “Applying Guidance in a Limited Interval Improves Sample and Distribution Quality in Diffusion Models”. *NeurIPS* (2024)

Summary

- The authors show that guidance is clearly harmful toward the beginning of the backward process (high noise levels) leading towards a handful template images per prompt greatly reducing variation, largely unnecessary toward the end (low noise levels), and only beneficial in the middle as it causes samples with more decisive set of features with crisper and perceptually more pleasing results.

- Limiting the guidance interval allows the use of much higher guidance weight.

- The proposed guidance interval is based on a piecewise-linear function, which sets w to 1 outside the guidance interval, removing the influence of the unconditional model. Smoother weighing functions did not improve results. Disabling guidance at selected sampling steps also did not work, guidance seems to have a cumulative effect over multiple consecutive steps.

Questions

- The authors claim to estimate the optimal guidance interval in practice, the lower limit should be set to zero first to optimize the upper limit, then fix the upper limit and optimize the lower limit. However, they wrote in the introduction that limiting the guidance interval allows for a higher guidance weight. So it seems counterintuitive that one can have a constant weight but with difference sized intervals.

7 Understanding Generalizability of Diffusion Models Requires Rethinking the Hidden Gaussian Structure

Xiang Li et al. "Understanding Generalizability of Diffusion Models Requires Rethinking the Hidden Gaussian Structure". *NeurIPS* (2024)

Summary

- In previous work, it was shown that diffusion models exhibit good generalization capacity. During training, recent research demonstrates that diffusion models operate in two distinct regimes: (i) a memorization regime, where models primarily reproduce training samples and (ii) a generalization regime, where models generate high-quality, novel images that extend beyond the training data. In the generalization regime, a particularly intriguing phenomenon is that diffusion models trained on non-overlapping datasets can generate nearly identical samples.

- The authors show that diffusion models have an inductive bias towards capturing and utilizing the Gaussian structure (covariance information) of the training set. This inductive bias seems unique to diffusion models and becomes more apparent when the model's capacity is relatively small compared to the training data set size. In the overparameterized regime, this inductive bias emerges during initial training phases before the model fully memorizes the training data.

- The recently observed strong generalization results from diffusion models learning certain common low-dimensional structural features shared across non-overlapping datasets can be partially explained through the Gaussian structure.

- The observed Gaussian structure is the optimal solution to the denoising score matching objective under the constraint that the model is linear.

- Empirically by distilling models from a non-linear model, in the high-noise regime of training, only coarse image structures are generated. The linear, Gaussian, multi-delta and non-linear model counterparts behave very similar. In the low-noise regime, only subtle, imperceptible details are added to the generated images. Both linear and multivariate Gaussian models effectively approximate the non-linear model. Even the nonlinear model exhibits high linearity shown by approximating an identity function with 1 on the skip connections and 0 for the network weights. In the intermediate-noise regime, realistic image content is generated with significant nonlinearity.

- A linear diffusion model can still generate images that resemble those generated with a nonlinear model in terms of overall image structure and some details.

- Modeling $p_{\text{data}}(x)$ as a multi-delta distribution reveals a key insight: while unconstrained optimal denoisers perfectly capture the scores of the empirical distribution, they have no generalizability. In contrast, Gaussian denoisers, despite having higher score approximation errors due to the linear constraint, can generate novel images that closely match those produced by the actual diffusion models. This suggests that the generative power of diffusion models stems from the imperfect learning of the score functions of the empirical distribution.

- Although real-world image distributions are significantly different from Gaussian, the findings imply that diffusion models have the bias towards learning and utilizing low-dimensional data structures, such as the data covariance, for image generation. However, the underlying mechanism by which the nonlinear diffusion models, trained with gradient descent, exhibit such linearity remains unclear and warrants further investigation.

- The authors define a generalization score to measure how well a diffusion model generalizes beyond the training data set:

$$\text{GL Score} := \frac{1}{k}\sum_{i=1}^{k}\frac{\left\|\boldsymbol{x}_i - \mathrm{NN}_Y\left(\boldsymbol{x}_i\right)\right\|_2}{\left\|\boldsymbol{x}_i\right\|_2}$$

Questions

- If it is true that during high-noise and low-noise regimes of diffusion sampling, the model shows high linearity, what does that mean for CFG? How does it behave differently? Is it connected to the CFG in the limited interval where CFG is only applied on the non-linear parts of sampling?
- What would happen if we were to apply the GL score to CFG? It should behave proportional to the guidance scale.

8 Boosting Diffusion Models with Moving Average Sampling in Frequency Domain

CVPR. "Boosting Diffusion Models with Moving Average Sampling in Frequency Domain" (2024)

Summary

- The authors propose to use moving averages over the x_0-prediction to stabilize sampling. For DDIM, this is straightforward plug-and-play. For other samplers, the x_0-prediction can be calculated by the noise prediction and then plugged in for sampling.
- Through the intuition that the diffusion model first generates low-frequency signals and then gradually adds high-frequency details, they employ a Discrete Wavelet Transform (DTW) to decompose the x_0-prediction into four wavelet subbands and separately apply moving averages with different parameters on each band. At each time step, the bands are converted back to the image domain via Inverse DWT.

Questions

- The method is only evaluated on FID, the diversity is not measured so the results are difficult to interpret.

9 Characteristic Guidance: Non-linear Correction for Diffusion Model at Large Guidance Scale

Candi Zheng and Yuan Lan. "Characteristic guidance: Non-linear correction for diffusion model at large guidance scale". *ICML* (2023)

Summary

- The original CFG formulation relies on the mixing of scores for the conditional and unconditional model, but the authors say this only holds for $t = 0$. The authors propose a mixing error, which measures the deviation of the guidance method from the Fokker-Planck equation (the learned score function is a solution of the score FP equation, the deviation from this leads to errors in sampling due to removing the equivalence between forward and backward diffusion process). For CFG, the mixing error comes from the non-linear terms of the FP equation, where the amplitude escalates with an increase in guidance scale and intensifies as the variance decreases.
- The paper proposes characteristic guidance as a systematic solution to the large guidance scale issue by addressing the nonlinearity neglicted in classifier-free guidance. Characteristic guidance employs a correction term with a projection operation. This projection differences between pixel space diffusion, latent space diffusion and low dimensional cases.
- The non-linear corrections predominantly occur in the middle stages of the diffusion process.

Questions

- The authors claim that CFG suffers from severe bias and catastrophic loss of diversity when ODE based sampling methods (ODE, DDIM, and DPM++2M) are used. Why is that the case?
- The characteristic guidances effect on latent space diffusion models (LDM) (Rombach et al., 2021) is different: it enhances the control strength rather than lowering the FID. It may simply be the choice of projection but its not clear why.

10 REG: Rectified Gradient Guidance for Conditional Diffusion Models

Zhengqi Gao et al. "REG: Rectified Gradient Guidance for Conditional Diffusion Models". *ICML* (2025)

Summary

- The authors claim that the currently employed CFG is only an approximating to the correct time-aware CFG, which would in practice prohibitively require sampling x_0 for each x_t.
- The paper proposes a rectified guidance term. Despite lower error and better FID/IS pareto fronts, it requires computation of an additional Jacobian matrix compared to vanilla CFG.

11 Eliminating Oversaturation and Artifacts of High Guidance Scales in Diffusion Models

Seyedmorteza Sadat et al. "Eliminating oversaturation and artifacts of high guidance scales in diffusion models" (2025)

Summary

- The authors disentangle the CFG update term into parallel and orthogonal components and show that the parallel component primarily causes oversaturation, while the orthogonal component enhances image quality.
- Rescaling and momentum is added to the CFG updates by the gradient ascent analogy. The rescaling is to control large update norms, which can cause drifts in the sampling process, they propose to constrain the updates within a sphere. The momentum is used in a repulsive way to down-weigh components arelady present in previous sampling steps, they call it reverse momentum.
- Adaptive projected guidance (APG) retains the quality-boosting advantages of CFG while preventing oversaturation at higher guidance scales.

Questions

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12 Overall Summary and Questions

- CFG is a technique to mix scores of a conditional and unconditional model to balance mode coverage and sample fidelity. This combined model is strangely better than a conditional or an unconditional model. In practice, the conditional model gets the majority of the compute budget and its score is most influential during sampling (high weight).
- CFG was originally intended to construct a new noise network to sample from a scaled distribution instead. This has been shown to not be the case for simpler tasks such as Gaussian mixtures. The intermediate distributions and the final distributions are very difficult to characterize outside specifically constructed cases (multi-delta, Gaussian, linear model).
- CFG provides a way to trade-off mode coverage (prompt alignment, diversity) with sample fidelity, although empirically it has been shown that very high prompt alignment and very high sample fidelity can be achieved while maintaining high generalization capability and only sacrificing some diversity in generations. Why and how this happens is still unclear.

- One intuition why CFG works so well could be that we can not train but we can sample from a higher quality underlying data distribution. Natural images contain for example many objects and are often cluttered, whereas there is a subset of clean images with a single object which is perceived as higher quality. It is however questionable how to measure the higher perceived quality (FID compares to samples from the distribution).
- Since people are maximizing the FID and IS scores as this is measurable, are there optimal mixtures of the conditional / unconditional scores such that each metric is optimized?
- CG takes an unconditional model and includes a gradient ascent step towards a classifier. CFG largely takes a conditional model and removes "unconditional likelihood". Could we go back and do the same for CG, so train a conditional model and then during sampling increase diversity through increasing classifier entropy?

13 TODOs

- Gabriel Raya and Luca Ambrogioni. "Spontaneous symmetry breaking in generative diffusion models". *NeurIPS* (2023)
- Arpit Bansal et al. "Universal Guidance for Diffusion Models". *CVPR Workshops* (June 2023)
- Yuchen Wu et al. "Theoretical insights for diffusion guidance: a case study for Gaussian mixture models". *ICML* (2024)
- Xi Wang et al. "Analysis of classifier-free guidance weight schedulers". *TMLR* (2024)
- Binxu Wang and John Vastola. "The Unreasonable Effectiveness of Gaussian Score Approximation for Diffusion Models and its Applications". *TMLR* (2024)
- Seyedmorteza Sadat et al. "CADS: Unleashing the Diversity of Diffusion Models through Condition-Annealed Sampling". *ICLR* (2024)
- Yifei Shen et al. "Understanding and improving training-free loss-based diffusion guidance". *NeurIPS* (2024)
- Chen Chen et al. "Towards memorization-free diffusion models". *CVPR* (2024)
- Marta Skreta et al. "The Superposition of Diffusion Models Using the It\^ o Density Estimator". *ICLR* (2025)
- Seyedmorteza Sadat et al. "No training, no problem: Rethinking classifier-free guidance for diffusion models". *ICLR* (2025)

- Chubin Chen et al. “S^2-Guidance: Stochastic Self Guidance for Training-Free Enhancement of Diffusion Models”. *arXiv 2508.12880* (2025)
- Krunoslav Lehman Pavasovic et al. “Classifier-Free Guidance: From High-Dimensional Analysis to Generalized Guidance Forms”. *arXiv* (2025)
- Jie Shao et al. “Images Speak Louder Than Scores: Failure Mode Escape for Enhancing Generative Quality”. *arXiv preprint arXiv:2508.09598* (2025)
- Marta Skreta et al. “Feynman-kac correctors in diffusion: Annealing, guidance, and product of experts”. *ICML* (2025)
- Peng Sun et al. “Unified Continuous Generative Models”. *arXiv preprint arXiv:2505.07447* (2025)
- Valentin De Bortoli et al. “Distributional diffusion models with scoring rules”. *ICML* (2025)
- Qiao Sun et al. “Is Noise Conditioning Necessary for Denoising Generative Models?” *arXiv preprint arXiv:2502.13129* (2025)
- Zhicong Tang et al. “Diffusion models without classifier-free guidance”. *arXiv preprint arXiv:2502.12154* (2025)
- Zhen Yang et al. “Efficient Training-Free High-Resolution Synthesis with Energy Rectification in Diffusion Models”. *arXiv preprint arXiv:2503.02537* (2025)
- Xiaoming Zhao and Alexander G Schwing. “Studying Classifier (-Free) Guidance From a Classifier-Centric Perspective”. *arXiv preprint arXiv:2503.10638* (2025)
- Rafał Karczewski et al. “Devil is in the details: Density guidance for detail-aware generation with flow models”. *ICML* (2025)
- Kulin Shah et al. “Does generation require memorization? creative diffusion models using ambient diffusion”. *ICML* (2025)